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**Activity Report
on**

“Study on Photodetector”

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(Savitribai Phule Pune University)



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Title: Study on Photodetector

Aim: To study the structure, working, and characteristics of a photodetector, with a focus on the photodiode.

Introduction :

A photodetector is a device that senses light and converts it into an electrical signal. These devices are essential in fiber optic communication systems where they detect the light signals transmitted through optical fibers and convert them back to electrical form for processing. Photodetectors are used in a wide range of applications, from simple light sensors in household devices to highly sensitive detectors used in scientific and industrial equipment.

One of the most common types of photodetectors is the **photodiode**. It is a type of semiconductor diode that operates under reverse bias and generates a photocurrent when exposed to light. The performance of a photodetector depends on its sensitivity, response time, spectral response, and noise characteristics.

Photodiodes are made from materials such as **Silicon (Si)**, **Germanium (Ge)**, or **Indium Gallium Arsenide (InGaAs)**, depending on the wavelength of light to be detected. They are widely used because of their **high speed, compact size, low power consumption**, and ability to be **integrated with electronic circuits**.



Fig : - Photodetector

Types of Photodetectors:

1. Photodiodes
2. Phototransistors
3. Avalanche Photodiodes (APD)
4. PIN Photodiodes
5. Photoresistors (LDRs)

Among these, the **photodetector** is widely used in optical communication due to its fast response and sensitivity to light.

Structure of a Photodiode :

The **photodetector**, especially the **photodiode**, is a semiconductor device that converts light into an electrical current. Its structure is designed to allow maximum light absorption and efficient electron-hole pair generation.

Key Layers Explained:

1. **P-Type Semiconductor:** This is doped with acceptor atoms to create holes as majority carriers.
2. **N-Type Semiconductor:** Doped with donor atoms to create electrons as majority carriers.
3. **Depletion Region:** When the P and N materials are joined, a depletion region forms at the junction. This is where electron-hole pairs are generated when photons strike the surface.
4. **Transparent Window/Coating:** Allows light to reach the depletion region without significant absorption or reflection.
5. **Metal Contact Layer:** Provides electrical connections to the external circuit.
6. **Substrate/Base:** Provides mechanical support.

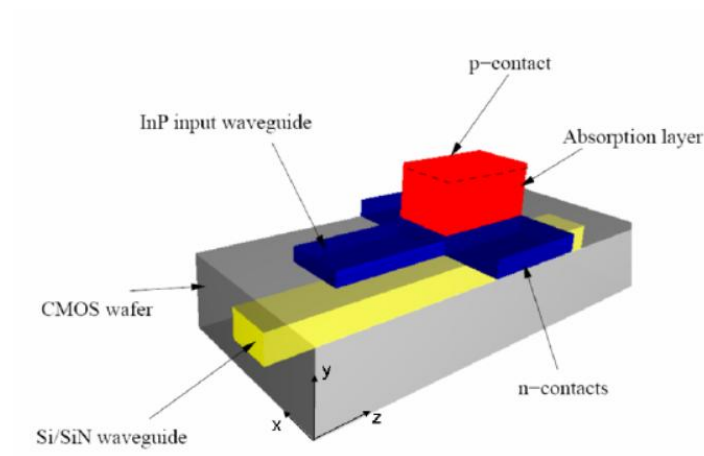


Fig : - Structure of Photodetector

Working Principle:

The photodiode works under reverse bias conditions. When photons strike the depletion region of the photodiode, they generate electron-hole pairs. The electric field across the junction causes these charge carriers to move, generating a photocurrent. The amount of current produced is proportional to the intensity of light falling on the diode.

The working of a **photodiode-based photodetector** is based on the **photoelectric effect**, where light energy is converted into electrical energy.

1. Structure Behavior Under Reverse Bias

When a photodiode is connected in **reverse bias**, the P-N junction's depletion region widens. In this state:

- There is minimal current flow in the absence of light (called **dark current**).
- The electric field across the depletion region is strong enough to sweep away charge carriers quickly.

2. Light Absorption and Electron-Hole Pair Generation

When incident photons (light particles) strike the surface of the photodiode:

- If the **photon energy** ($E = hf$) is greater than or equal to the **bandgap energy** of the semiconductor material, it excites electrons from the **valence band to the conduction band**, creating **electron-hole pairs**.
- This generation typically happens in the **depletion region**, where the electric field is present.

3. Carrier Separation and Photocurrent Generation

The electric field across the depletion region causes:

- **Electrons** to be swept towards the **N-side**.
- **Holes** to be swept towards the **P-side**.

This movement of charge carriers constitutes a **photocurrent**, which is directly **proportional to the intensity of incident light**.

4. Output Signal

The current is collected at the terminals of the photodiode, producing a signal that can be:

- Measured directly as current.
- Converted into voltage using amplifying circuits.
- Digitally processed for further interpretation.

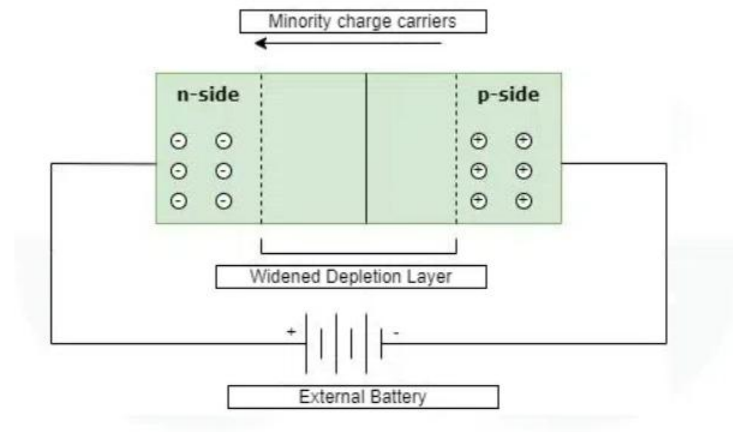


Fig : - Working of Photodetector

Working Steps:

1. Apply reverse bias voltage to the photodiode.
2. Light enters through the transparent window and hits the depletion region.
3. Photons generate electron-hole pairs.
4. Electrons move toward the N-side and holes toward the P-side.
5. The movement creates a current that is measured as the output.

Advantages of Photodiodes:

- High sensitivity to light
- Fast response time
- Compact and lightweight
- Compatible with digital circuits
- Low power consumption

Disadvantages of Photodiodes:

- Requires external biasing
- Low output signal strength (requires amplification)
- Sensitive to temperature variations
- Narrow spectral response range (depending on material)

Applications:

- Fiber optic communication
- Optical switches
- Light meters
- Safety and security systems
- Medical instruments

Conclusion:

Photodetectors, especially photodiodes, play a critical role in converting optical signals to electrical signals in fiber optic communication. Their fast response time, reliability, and sensitivity make them ideal for such applications.

